

Polynomials: a root is a factor in disguise

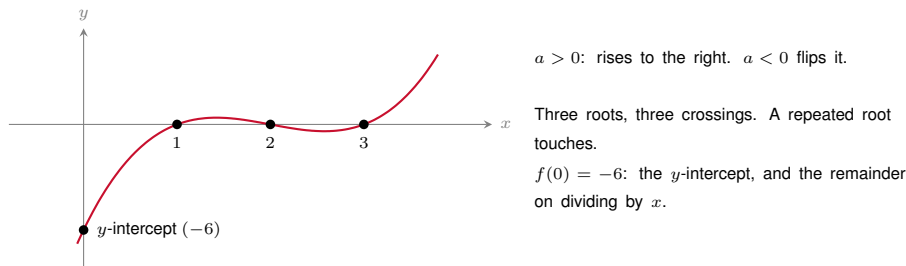
THE ONE IDEA

A root is a factor in disguise. If $f(a) = 0$ then $(x - a)$ divides $f(x)$ exactly; if not, the remainder is $f(a)$.

EXAMPLE: THE CUBIC THE NOTES KEEP RETURNING TO

$$y = \underbrace{x^3 - 6x^2 + 11x - 6}_{\text{standard form, degree 3}} = \underbrace{(x - 1)(x - 2)(x - 3)}_{\text{factored form: roots 1, 2, 3}}$$

THE PICTURE



FOUR TOOLS, FOUR FACTS

Tool	Statement	What it hands you	In the notes' examples
Long division	$f = gq + r$, $\deg r < \deg g$	The replay: quotient q and remainder r , step by step. $\deg q = \deg f - \deg g$.	$2x^3 - 4x^2 + 5x - 1$ by $x^2 + 3x + 1$: $q = 2x - 10$, $r = 33x + 9$.
Remainder theorem	$f(x) = (x - a)q(x) + f(a)$	The scoreline without the replay: dividing by $(x - a)$ leaves $f(a)$. Dividing by $(x + 2)$ means $a = -2$.	$f = x^3 + x^2 - x + 2$: $f(1) = 3$, $f(-1) = 3$, $f(2) = 12$, $f(-2) = 0$.
Factor theorem	$f(a) = 0 \iff (x - a) \mid f(x)$	A root is a factor. Given one root, divide by its factor and solve the quadratic that is left.	$f(-2) = 0$, so $x^3 + x^2 - x + 2 = (x + 2)(x^2 - x + 1)$; the quadratic has $\Delta < 0$, so one real root only.
Vieta	$S = -a_{n-1}/a_n$, $P = (-1)^n a_0/a_n$	The roots tell on each other: their sum and product sit in the coefficients before anyone finds them.	$2x^3 - 12x^2 + 22x - 12$, roots 1, 2, 3: $S = 6 = -\frac{-12}{2}$, $P = 6 = \frac{(-1)^3(-12)}{2}$.

THE METHOD HE WILL USE: ONE ROOT GIVEN, DIVIDE, FINISH WITH THE QUADRATIC

Example 7: $x^3 - 6x^2 + 11x - 6$ with $x = 1$ a root. Divide by $(x - 1)$: quotient $x^2 - 5x + 6 = (x - 2)(x - 3)$. Three roots. Example 6: $x^3 + x^2 - x + 2$ with $x = -2$ a root: quotient $x^2 - x + 1$, $\Delta = -3$, so the cubic has one real root and two the real numbers cannot see (Thursday).

VIETA FOR A QUADRATIC, WHICH IS WHERE IT COMES FROM

$ax^2 + bx + c = a(x - r_1)(x - r_2) = ax^2 - a(r_1 + r_2)x + ar_1r_2$, so $r_1 + r_2 = -\frac{b}{a}$ and $r_1r_2 = \frac{c}{a}$. For a cubic the same expansion gives $S = -\frac{b}{a}$ and $P = -\frac{d}{a}$. HL question type (set, Q15): $ax^3 + 6x^2 - 8x + 9$ with sum of roots 2 gives $-\frac{6}{a} = 2$, $a = -3$, product 3.

IN PLAIN ENGLISH

Divide a polynomial by $(x - a)$ and what is left over is just the height of the curve at a . Height zero means it divides exactly, and then a is a root and $(x - a)$ is a factor: two names for one fact.

TRAPS

- The sign: dividing by $(x + 2)$ means evaluating $f(-2)$, not $f(2)$.
- The leading coefficient: $2x^3 - 12x^2 + 22x - 12 = 2(x - 1)(x - 2)(x - 3)$, not $(x - 1)(x - 2)(x - 3)$.
- Vieta's product carries $(-1)^n$: for a cubic the product is $-\frac{d}{a}$.
- Degrees: $\deg(fg) = \deg f + \deg g$, but $\deg(f + g)$ can drop if leading terms cancel.

LISTEN FOR THESE THREE MOMENTS

- Which long-division layout he uses, left-to-right or right-to-left (the notes show both). Copy his.
- Which theorem he proves on the board, remainder or factor. The proof is three lines and is the same for both.
- Vieta for the cubic: does he expand $a(x - r_1)(x - r_2)(x - r_3)$ or state the formula? If he expands, that is the proof.

Questions to ask in class

Ask at least one. The first is safe, the second shows you followed, the third is the one you actually want answered. Write down what was said; it is the part you will remember.

1. *[safe]*

When we divide by $(x + 2)$, is the remainder $f(-2)$ or $f(2)$, and why does the sign flip?

What you expect: $f(-2)$: the factor is $x - a$ with $a = -2$. Gets the one trap the whole class will fall into said out loud.

What he said:

2. *[shows you followed]*

In Example 6 the quadratic factor has $\Delta < 0$. So the cubic has one real root. Does it still have three roots in some sense, and will we count them that way?

What you expect: Yes: two complex roots, a conjugate pair. He will either say so now or on 14 October; either way it links 2.11 to 1.12.

What he said:

3. *[the one you want answered]*

Vieta says the sum and product of the roots are fixed by the coefficients. Does that hold when some roots are complex, and is that why complex roots of a real polynomial must come in conjugate pairs?

What you expect: It holds over the complex numbers, and a real sum and real product force the pair. Expect him to answer with the quadratic formula first.

What he said:

Later. Next HL hour (14 October): Vieta for higher degree and the HL question types. Next HL section: rational functions, where the remainder refuses to go away.

Next. Thursday's double: shared class, with the third episode of the week (the numbers nobody wanted). Sprint 2 is due Thursday 20:00.